

PIONEER JUNIOR COLLEGE

JC2 PRELIMINARY EXAMINATION
HIGHER 2

CANDIDATE
NAME

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CT
GROUP

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INDEX
NUMBER

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CHEMISTRY

9729/02

Paper 2 Structured

12 September 2017

2 hours

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your class, index number and name on all work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough workings.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINER'S USE			
Paper 1		/ 30	
Paper 2			
1	/ 14	4	/ 12
2	/ 8	5	/ 13
3	/ 11	6	/ 17
Penalty		sf	unit
Paper 2		/ 75	
Paper 3		/ 80	
Paper 4		/ 55	
Total			

Answer **all** questions in the spaces provided.

- 1 (a)** Carbon dioxide, produced mainly from the burning of fossil fuels, is a greenhouse gas which trapped heat within the Earth's atmosphere, causing global warming and potentially catastrophic global climate change. One major source of carbon dioxide produced worldwide is from the rapidly expanding aviation industry.

- (i)** Commercial passenger jet-liners combust kerosene as fuel. The formula of kerosene may be taken as $C_{14}H_{30}$. Write a balanced equation, with state symbols, for the complete combustion of kerosene.

..... [1]

In 2017, Singapore Airlines announced that it will be launching the world's longest commercial non-stop passenger flight from Singapore to New York, with a flight distance of 15,000 km. The ultra-long range Airbus A350 aircraft used on this route burns 7.5 kg of kerosene per km.

- (ii)** Calculate the mass of carbon dioxide produced, in tonnes, during this flight.
[1 tonne = 1000kg]

[3]

- (iii)** Apart from carbon dioxide, nitrogen dioxide is another air pollutant found in the flue gases from the jet engine of aircrafts. Explain how nitrogen dioxide is produced in the jet engine.

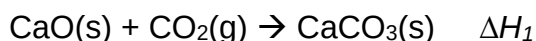
.....
..... [1]

- (iv)** Why is it important to remove nitrogen dioxide from the flue gases?

.....
..... [1]

- (b)** Scientists have been studying the idea of carbon capture and storage technologies

to try to remove atmospheric carbon dioxide on an industrial scale. One promising technology is called *calcium looping*. This process involved using calcium oxide to react with the atmospheric carbon dioxide to form calcium carbonate (i.e. limestone) which can then be stored away in abandoned mines.



- (i) Using the space below, construct a fully labelled energy cycle diagram to calculate ΔH_1 using the following data.

standard enthalpy change of formation of CaO(s)	-635 kJ mol ⁻¹
standard enthalpy change of formation of CaCO ₃ (s)	-1207 kJ mol ⁻¹
standard enthalpy change of combustion of C(s)	-394 kJ mol ⁻¹

[3]

- (ii) Like calcium carbonate, aluminum oxide is a major components of naturally occurring rocks. Explain clearly, in terms of bonding and structure, why aluminum oxide is insoluble in water.

.....

.....

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.....

.....

..... [3]

- (c) Carbonate buffers are widely used for biochemical purposes.

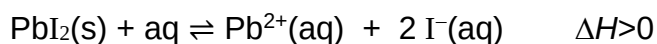
A student wishes to form a $\text{CO}_3^{2-}/\text{HCO}_3^-$ buffer in the laboratory.

Given that the carbonate ion has a K_b of $1.8 \times 10^{-4} \text{ mol dm}^{-3}$, calculate the volume of 0.10 mol dm^{-3} of hydrochloric acid that he has to add to 25.0 cm^3 of 0.12 mol dm^{-3} of Na_2CO_3 , to form a resulting buffer solution of pH 9.50.

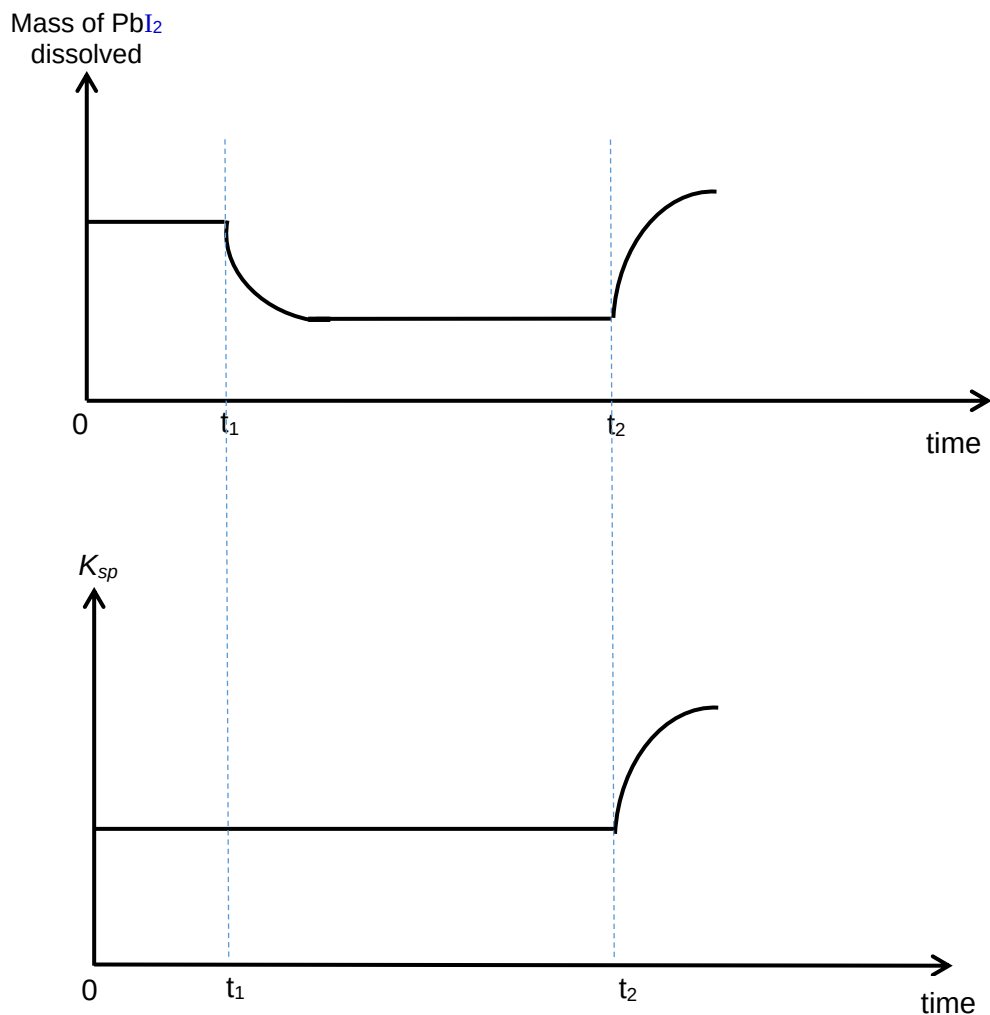
[2]

[Total:14]

2 (a) This question is about lead(II) halides. PbI_2 is a sparingly soluble salt:



A student aimed to investigate the solubility product and solubility of PbI_2 under different conditions. He first added some solid PbI_2 into 100 cm^3 of water at 15°C , and subsequently made two changes at t_1 and t_2 . The graphs below show the mass of PbI_2 dissolved and solubility product of PbI_2 against time.



Some possible changes listed below were made by the student at t_1 and t_2 .

1. Cool reaction mixture to 10°C in water bath
2. Heat reaction mixture to 50°C in water bath
3. Add $\text{AgNO}_3(\text{s})$
4. Pump in $\text{Pb}(\text{NO}_3)_2(\text{aq})$

With reference to **both** graphs, suggest and explain which of the above changes (1)-(4), was made at t_1 and t_2 respectively.

change made at t_1 :

explanation:.....

.....[2]

change made at t_2 :

explanation:.....

[2]

- (b) The values of the solubility products of two salts, lead(II) iodide, PbI_2 , and silver (I) iodide, AgI are given below.

$$K_{\text{sp}} (\text{PbI}_2) = 3.97 \times 10^{-9} \text{ mol}^3 \text{ dm}^{-9}$$

$$K_{\text{sp}} (\text{AgI}) = 8.3 \times 10^{-17} \text{ mol}^2 \text{ dm}^{-6}$$

A lab technician added solid PbI_2 into water, shaking it thoroughly. He filtered off the remaining solid, leaving a saturated solution **H**.

He added drops of aqueous silver nitrate to **H**, until AgI just precipitates. Determine the concentration of Ag^+ when AgI just precipitates.

[2]

- (c) When a precipitate is formed, $\Delta G^\circ_{\text{ppt}}$, in J mol^{-1} , is given by the following expression.

$$\Delta G^\circ_{\text{ppt}} = 2.303 RT \log K_{\text{sp}}$$

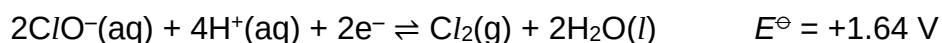
Given that for lead(II) chloride, PbCl_2 , $K_{\text{sp}} = 1.70 \times 10^{-5} \text{ mol}^3 \text{ dm}^{-9}$ at 298 K, calculate $\Delta G^\circ_{\text{ppt}}$ for PbCl_2 at 298K. Thus explain if PbCl_2 is soluble at 298K.

[2]
 [Total:8]

- 3 (a) Chlorine has been extensively used in the manufacture of many compounds.

One such compound is household bleach, which contains sodium chlorate(I), NaClO .

The chlorate(I) ion acts as a powerful oxidising agent in acidic solution:



(i) Define the term *standard electrode potential*.

.....
[1]

(ii) Draw a fully labelled diagram, to show how the standard electrode potential, $E^\ominus(\text{ClO}^-/\text{Cl}_2)$, could be measured.

[3]

(b) Use the *Data booklet* to predict the reaction, if any, that would occur when the following pairs of reagents are mixed in acidified aqueous solution. If a reaction occurs, write a balanced equation and calculate $E^\ominus_{\text{cell}}$.

(i) $\text{NaClO}(\text{aq})$ and $\text{H}_2\text{O}_2(\text{aq})$

[2]

(ii) $\text{NaClO}(\text{aq})$ and $\text{Cr}(\text{NO}_3)_3(\text{aq})$

[2]

- (c) When a few drops of aqueous sodium carbonate was added to chromium(III) nitrate, carbon dioxide gas and a grey-green precipitate, **R**, were obtained.

With the aid of equation(s), explain why CO_2 is given off in the above reaction, and identify precipitate **R**.

.....

.....

.....

.....

.....

..... [3]

[Total:11]

- 4 Manganese is a first row d-block element in the Periodic table with a melting point of 1246°C . Like many other d-block elements, manganese shows properties that are atypical

from other s-block metals, such as potassium and calcium.

Its minerals are widely distributed, with manganese dioxide and manganese carbonate being the most common. Manganese is also an essential element in living organisms, where many types of enzymes make use of the variable oxidation states of manganese to take part in the various redox reactions.

- (a) (i) Explain why manganese has a much higher melting point than calcium, which is found in the same period.

.....

 [3]

- (ii) Explain briefly why the density of manganese is significantly greater than that of calcium.

.....
 [1]

- (b) Heating a mixture of KOH and MnO_2 in air gives potassium manganite, K_2MnO_4 , which is an important precursor to potassium permanganate, KMnO_4 , a common oxidising agent.

- (i) State the oxidation state of manganese and hence, write the electronic configuration of the manganese ion in each of the following compounds:

MnO_2 :

Oxidation state:

Electronic configuration:

K_2MnO_4 :

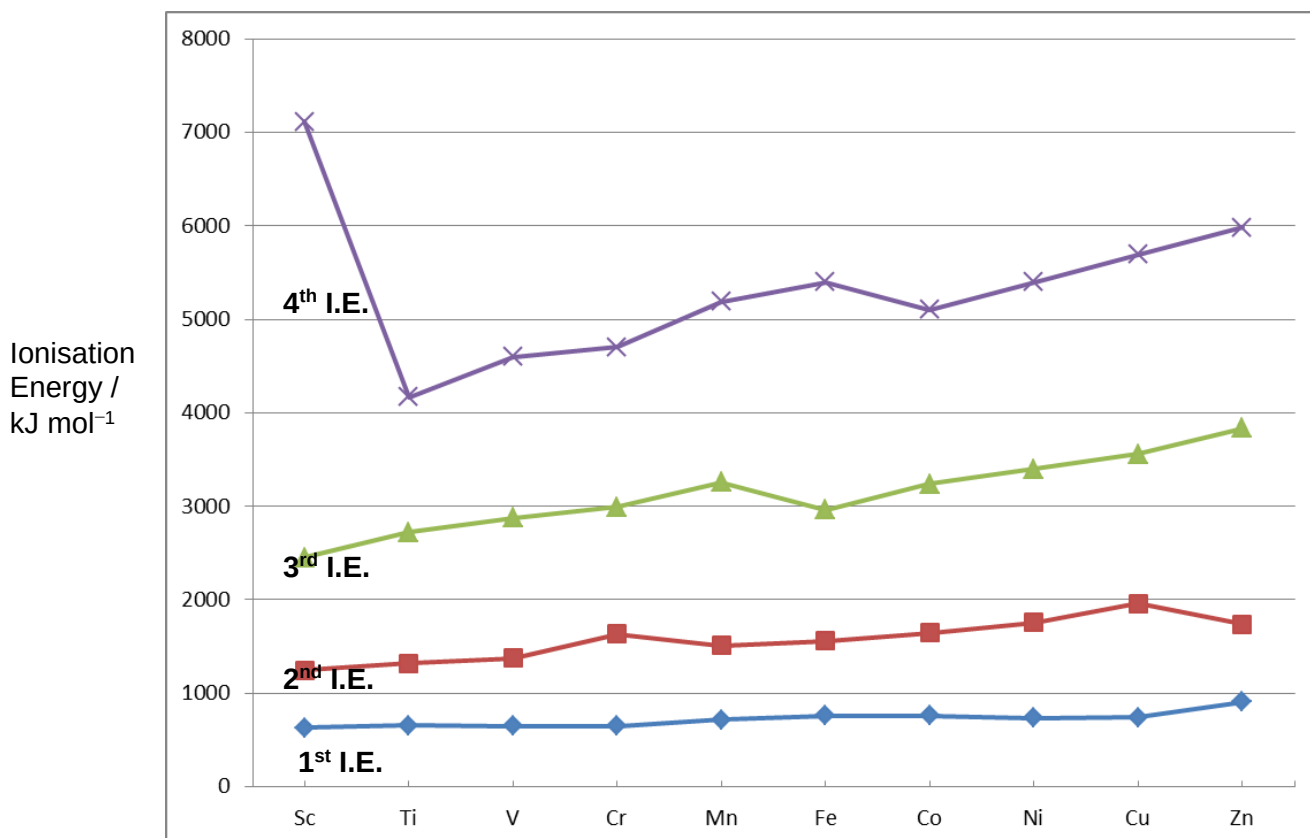
Oxidation state:

Electronic configuration: [2]

- (ii) Write a balanced equation for the redox reaction between MnO_2 , KOH and air.

.....[1]

- (c) The graph below shows the variation of the first to fourth ionisation energies for the first row d-block elements scandium to zinc.



- (i) Give an equation that represents the second ionisation energy of manganese.

.....[1]

- (ii) Explain why the first ionisation energies of the elements Ti to Cu, are relatively invariant across the Period.

.....

[2]

- (iii) Explain why the fourth ionisation energy of cobalt is lower than that of iron.

.....

[2]

[Total:12]

- 5 (a) Structure elucidation is an important skill that chemists must acquire to help them determine the chemical structures of organic compounds.

A novice chemist requires your assistance in determining the structure of an aromatic organic compound **P**, that she found in the laboratory. The molecular formula of compound **P** is $C_9H_{11}ON$, and **P** contains two functional groups.

- (i) An orange precipitate is observed when **P** is reacted with 2,4-dinitrophenylhydrazine. Name the type of reaction that occurred. From this observation only, name two different functional groups that could be present in **P**.

Type of reaction:

Functional groups: and

[2]

- (ii) No silver mirror is formed when **P** is warmed with hot ammoniacal silver nitrate (mixture of aqueous ammonia and aqueous silver nitrate). Which of the functional groups you named in (a)(i), is confirmed by this observation?

..... [1]

- (iii) When compound **P** is reacted with bromine water in excess, a white solid which has $M_r = 385.7$, is formed. Deduce the molecular formula of the white solid that is formed.

..... [1]

- (iv) Compound **P** reacts with hot potassium manganate(VII) in aqueous sodium hydroxide. A product with the molecular formula, $C_8H_5NO_4Na_2$, is formed. Draw the structure of the product.

[1]

- (v) You now have enough information to determine the structural formula of compound **P**. Draw the structure for **P**, explaining clearly why you have placed each of the two functional groups in their respective positions.

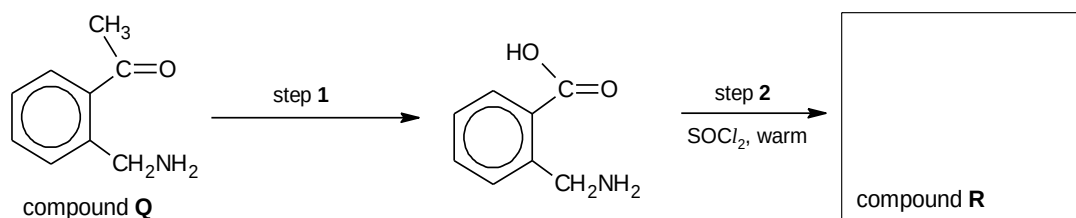
.....

[2]

(b) Compound **Q**, $C_9H_{11}ON$, is a structural isomer of compound **P**.

The diagram below shows a possible reaction scheme for converting compound **Q** to **R**, C_8H_7NO .

(i) Complete the diagram by suggesting the reagents and conditions for step 1, and give the structural formula of compound **R**.



[2]

(ii) It was found that the yield of **R** was low. Suggest a reason for this.

.....
 [1]

(c) The table below shows the pK_b values of two organic nitrogen compounds in aqueous solutions.

Compound	pK_b
methylamine, CH_3NH_2	3.36
trimethylamine, $(\text{CH}_3)_3\text{N}$	4.19

Using the data provided in the table above, suggest an explanation on the relative basicities of these two compounds.

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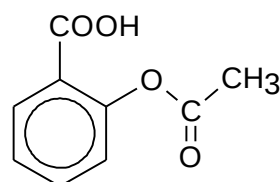
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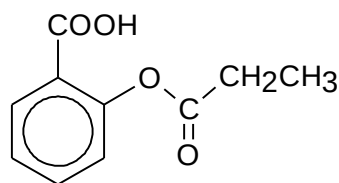
[Total:13]

- 6 (a) Acetylsalicylic acid (ASA), which is extracted from the leaves of the willow tree, has been used for its health effects for at least 2,400 years. It is an important medication that is used to treat pain, fever, and inflammation.



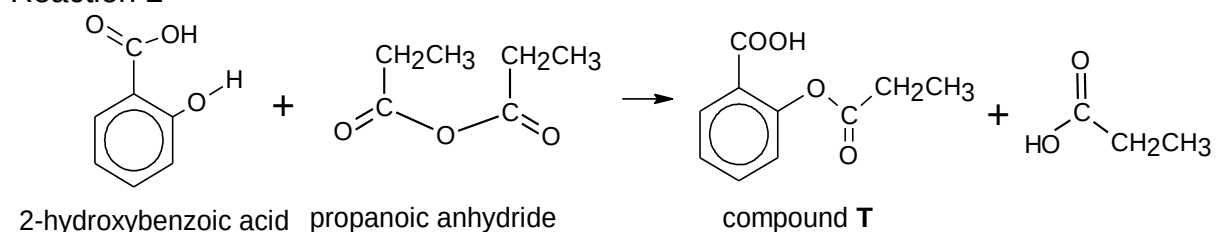
Acetylsalicylic acid

A chemist wanted to form a derivative of ASA, compound **T** below:

Compound **T**

One method to form compound **T** was to react 2-hydroxybenzoic acid and propanoic anhydride in reaction **1** below.

Reaction 1



(i) State the type of reaction that has occurred in reaction **1**.

.....[1]

The mechanism of reaction **1** consists of four steps and is described below:

Step 1:

The phenol group on 2-hydroxybenzoic acid acts as a nucleophile and attacks one of the reactive carbons in propanoic anhydride to form only one intermediate **W**. The intermediate **W** contains both a cationic oxygen and an anionic oxygen.

Step 2:

A proton is released from intermediate **W** to form intermediate **X**.

Step 3:

From intermediate **X**, a carboxylate anion is expelled as a leaving group, and an ester **Y** is formed as the other product.

Step 4:

Protonation of the carboxylate anion formed in step **3** produces a carboxylic acid.

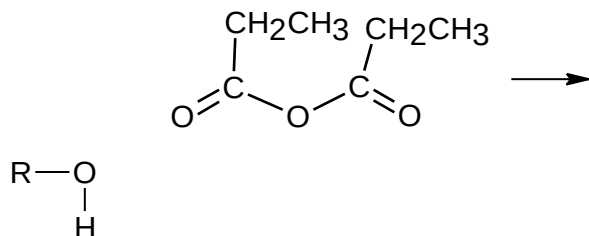
(ii) Draw all four steps in the full mechanism for the reaction between 2-hydroxybenzoic acid and propanoic anhydride.

(you may use ROH to represent 2-hydroxybenzoic acid in your answer.)

For the first step of the mechanism, *complete* the diagram shown below and draw the remaining three steps of the mechanism in the space given below. Show all intermediates, relevant charges, lone pairs of electrons and indicate the movement of electron pairs with curly arrows in **all** the steps.

Step 1

Let R—O—H represents 2-hydroxybenzoic acid.



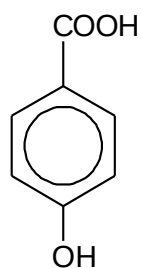
Step 2

Step 3

Step 4

[4]

(iii) An isomer of 2-hydroxybenzoic acid is shown below:



4-hydroxybenzoic acid

Both 2-hydroxybenzoic acid and 4-hydroxybenzoic acid are dibasic acids that ionises in two stages. The acid dissociation constant, pK_1 , for the first stage of dissociation is given below.

	pK_1
	4.54
	2.97

With the aid of a diagram, explain why 2-hydroxybenzoic acid has a lower pK_1 .

.....

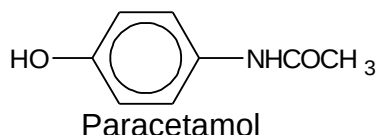
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.....

.....[2]

(b) Another medicine that is used widely to treat fever and pain is paracetamol, or

commonly known as panadol.



Some information on paracetamol is provided in Table 1 below.

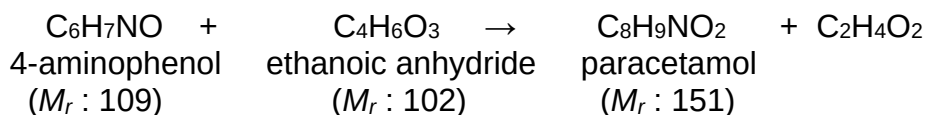
Data on paracetamol	
Molecular Formula	C ₈ H ₉ NO ₂
Density	1.263 g/cm ³
Melting Point	169°C
Boiling Point	420°C
Solubility in water	14 mg/ml
Bioavailability*	75% (oral) 100% (intravenous)
Biological half-life**	2 hours

Table 1

**Bioavailability refers to the percentage of a drug which enters the blood circulation when introduced into the body and hence able to have an active and effective effect.*

***Biological half-life of a drug is the time taken for the drug to reduce to half its original amount in the body.*

- (i) Paracetamol can be produced from the reaction between 4-aminophenol and ethanoic anhydride as shown below.



It is known that the yield of this reaction is only 60%.

Calculate the minimum masses of the reactants that are required to produce 10 g of paracetamol.

.....[2]

- (ii) A doctor wishes to prescribe oral medication for one of his patients suffering

from headache. In order for the medicine to be effective, the patient requires a recommended dosage of 560 mg of paracetamol in her body.

How many 250 mg paracetamol tablets should the patient take each time?

[2]

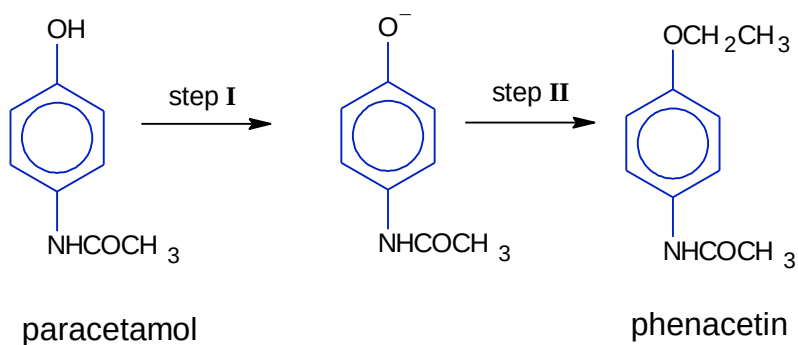
- (iii) A patient suffering from acute dental pain is advised to consume orally, four 250 mg paracetamol tablets.

To effectively alleviate his pain, at any point of time, the amount of paracetamol in his body **must not** drop below 93 mg.

With reference to relevant data from Table 1, and the use of suitable calculations, determine the maximum number of hours that the oral medication prescribed above remains effective, and show clearly whether the doctor should advise the patient to take four tablets of paracetamol once in every 4 hours, 6 hours, or 8 hours.

[2]

- (c) (i) Phenacetin was once used as an analgesic (pain killing drug). It can be synthesised from paracetamol below.



Suggest reagents and conditions for step I and II.

step I:

step II:

[2]

- (ii) A student added aqueous sulfuric acid to paracetamol and heated it. Give the structures of the products obtained.

[2]

[Total:17]

End of paper